VI_WCK

October 4, 2026

[4]:
```
include("notebook_init.jl")
```

This notebook runs the examples from the paper. We do the computations from a fixed point map view point. So if Ω is the reasible region and $\mathbf{F}$ is the operator, we seek the fixed point of the map

$$\mathbf{G}(x) = x - \Pi_\Omega(x - \tau\mathbf{F}(x))$$

where Π_Ω is the projection onto Ω. In each of the examples we will specify $\mathbf{F}$, τ, Ω, and an initial iteration x_0.

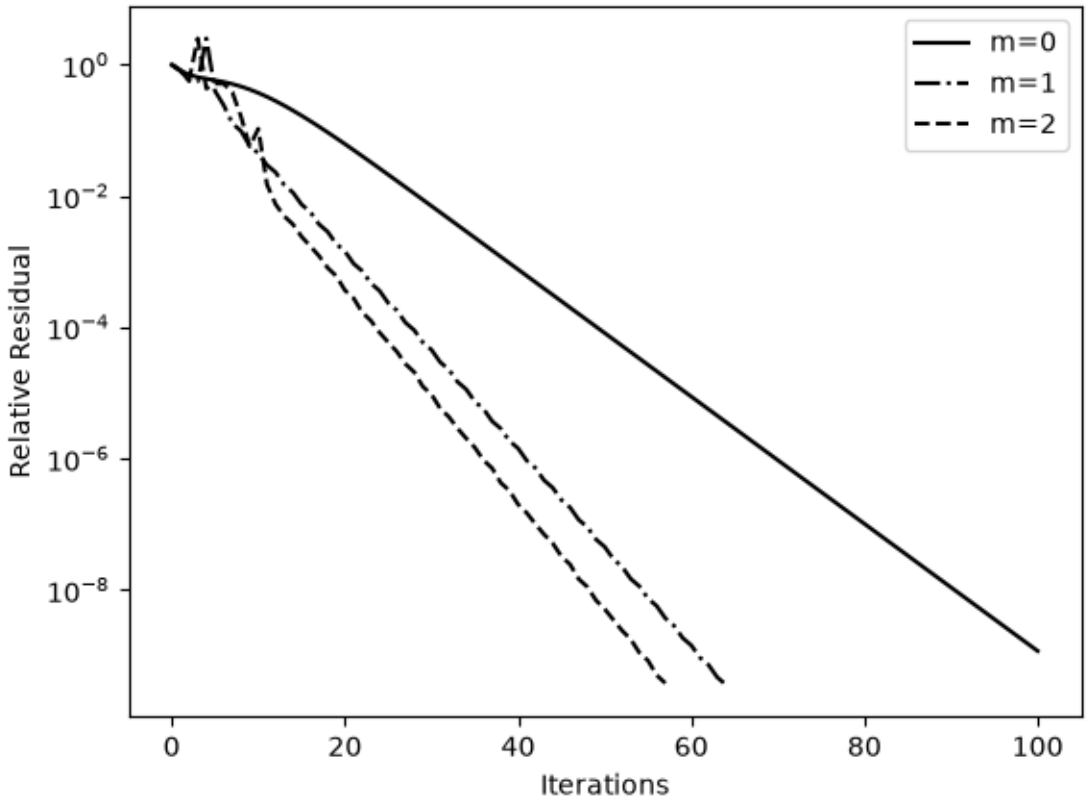
With this in mind, Alg 1 is simply a direct fixed point iteration for $\mathbf{G}$. We will compare that iteration Anderson Acceleration (AA) using the implementation from the Julia package SIAMFANLEquations.jl.

We begin with Algorthm 1 and example 2 from the paper. The data are $\tau = .1$ and $x_0 = (1, 2, -1)$. We show results from projected AA(m) for $m = 0, 1, 2$. AA(0) is the same as Alg 1 in the paper. I set a maximum iteration limit of 100 and the $m = 0$ case had not converged yet, hence the failure message.

The code is a projected version of the AA code from my book and the internal least squares method fails with a singularity for $m > 2$.

[5]:
```
AAE2Plot();
```

```
Failure to converge after 100 iterations for m=0 in aasol.jl
```


m=0
m=1
m=2
Relative Residual
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
0
20
40
60
80
100
Iterations

[ ]: